

Life Cycle Assessment (LCA) / ACV

What this tool produces · *A detailed environmental inventory and impact assessment report calculating embodied carbon, energy consumption, and material toxicity from extraction to end-of-life.*

G2 · Proof of Fit

G4 · Proof of Sustainability

Dimensions

Environmental

Economic

Preparing the workshop · *Life Cycle Assessment (LCA) / ACV*

G2 · Proof of Fit

Why this tool matters

Making the invisible supply chains of the city visible, revealing that every brick and steel beam carries a profound global history of extraction, labor, and atmospheric impact.

Evidence we produce

A detailed environmental inventory and impact assessment report calculating embodied carbon, energy consumption, and material toxicity from extraction to end-of-life.

Duration & Material

Duration · 4 to 8 weeks

Material · LCA software (e.g., SimaPro), Ecoinvent database, Environmental Product Declarations (EPDs)

Reference case

In Flanders, LCA was applied extensively to analyze the residential building stock. Researchers demonstrated that while low-energy houses excel in operational savings, ignoring the embodied carbon of their complex materials leads to problem-shifting. Incorporating cradle-to-grave LCA ensured that future building codes addressed the full ecological cost of construction and material disposal.

Suggested agenda · Brief (10 min) > Method walk-through (15 min) > Animation canvas (60-90 min) > Harvest (20 min)

Lifecycle stages · cradle → grave → cradle

Duration
4 to 8 weeks

Raw materials

extraction & sourcing

Manufacture

production · transport

Use phase

energy · maintenance

End-of-life

reuse · recycle · dispose

Beyond-life

second life · biosphere return

Quantify GWP, energy, water, biodiversity at each stage. Compare alternatives.

Harvesting our *learnings*

After running the canvas, capture three layers of insight.

What inspires me

Stand-out signals, surprising stories, ideas to remember

How to apply it here

Concrete adaptations to our context, our team, our constraints

What we need to be autonomous

Skills, allies, resources, decisions still to secure

Each participant fills a copy. Then debrief together to surface patterns and next steps.